1) 1 -> 999 distinct digits

Case 1: $1, 2, \ldots, 9$ — 9

Case 2: $\underline{\neq 0}\ \underline{\neq 4} = 81$

$\downarrow \quad \downarrow$

$9 \quad 9$

Case 3: $\underline{\ }\ \underline{\ }\ \underline{\ } = 9 \cdot 9 \cdot 8 = 648$

$\downarrow \quad \downarrow \quad \downarrow$

$9 \quad 9 \quad 8$

$\Rightarrow$ 738 nr ✓

2) ____

$\underline{\ }\ \underline{\ }\ \underline{\ }\ \underline{0} = 72 \cdot 7 = 504$

$9 \quad 8 \quad 7 \quad 1$

$\underline{\ }\ \underline{\ }\ \underline{\ }\ \underline{\neq 0} = 8 \cdot 8 \cdot 7 \cdot 4 = 64 \cdot 28 = 1792$

$\downarrow \quad \downarrow \quad \downarrow \quad \downarrow$

$8 \quad 8 \quad 7 \quad 4$

$\Rightarrow$ 2296 ✓

3) ____ — 9000 nr ~~total~~

$\underline{\ }\ \underline{\ }\ \underline{\ }\ \underline{\ } = 81 \cdot 56 = 4536$

$9 \quad 9 \quad 8 \quad 7$

$\Rightarrow$ 4464 ✓

4) $\underline{36}\ \underline{36}\ \underline{36}\ \underline{36}\ \underline{36}\ \underline{36}$

$\underline{\ }\ \underline{\ }\ \underline{\ }\ \underline{\ }\ \underline{\ }\ \underline{\ }$

$a, b, c, \ldots, z \mid = 36$

$0, 1, \ldots, 9$

$\underline{\ }\ \underline{\ }\ \underline{\ }\ \underline{\ }\ \underline{\ }\ \underline{\ }$

$26 \quad 26 \quad 26 \quad 26 \quad 26 \quad 26$

$36^6 - 26^6 = (36^3 + 26^3)(36^3 - 26^3)$

$(46656 + 17576) \ldots$ ✓

5) $1, 2, 3, \ldots, 20$ $\quad \overline{3}$

20

$= 20 \cdot 19 \cdot 18 = \ldots = 6840$ ✓

1	2	3
↓	↓	↓
A	19	18

$\Rightarrow 684 + 5814 = 6498$ ✓

1	2	3
↓	↓	↓
19	A	18

1	2	3
↓	↓	↓
19	18	17

1	↓	1
↓		
19	18	A

$\Rightarrow 6498$ ✓

$$\begin{array}{r} 27 \cdot \\ 27 \\ \hline 189 \\ 54 \\ \hline 729 \end{array}$$

6) $\underline{3}\ \underline{3}\ \underline{3}\ \underline{3}\ \underline{3}\ \underline{3} = 3^6 = 27 \cdot 27 = 729$ ✓

$\underline{2}\ \underline{2}\ \underline{2}\ \underline{2}\ \underline{2}\ \underline{2} = 2^6 = 64$

$\Rightarrow 729 - 64 = 665$ ✓

$\underline{3}\ \underline{2}\ \underline{2}\ \underline{2}\ \underline{2}\ \underline{2} = 2^5 \cdot 3 = 96$ ✓

7) $f: \{1, 2, 3, 4\} \to \{1, 2, 3, 4, 5, 6\}$

$6^4 = 36 \cdot 36 = 1296$ ✓

$$\begin{array}{r} 36 \cdot \\ 36 \\ \hline 216 \\ 108 \\ \hline 1296 \end{array}$$

$$6^{\underline{4}} = 6 \cdot 5 \cdot 4 \cdot 3 = 30 \cdot 12 = 360$$ ✓

$f(1) = 1 \to f(2) = 5$
$f(1) = 2 \Rightarrow f(2) = 4$
$f(1) = 3 \Rightarrow f(2) = 3$
$f(1) = 4 \Rightarrow f(2) = 2$
$f(1) = 5 \Rightarrow f(2) = 1$

$$= 15 \cdot \frac{6!}{6} = 36 \cdot 15 = 540$$ ✓

8) $\frac{n^{\underline{k}}}{n} = \frac{6^{\underline{c}}}{6} = 5! = 120$ ✓

